

Algorithmic Pseudocode Sample 55

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1: for variant in variants do
2:   dates  $\leftarrow$  unique dates in which variant exists
3:   dates.sort()
4:   model_initial_weights  $\leftarrow$  random
5:   for d in dates do
6:     retro_data  $\leftarrow$  all data before d
7:     processed_data  $\leftarrow$  preprocess retro_data
8:     dataset  $\leftarrow$  structured processed_data into graphs
9:     train_dataset, val_dataset  $\leftarrow$  temporally split dataset into 80% training and the most
recent 20% for validation
10:    model  $\leftarrow$  initialize desired model with model_initial_weights
11:    epochs  $\leftarrow$  100
12:    optimizer  $\leftarrow$  Adam(lr=0.05)
13:    early_stopper  $\leftarrow$  EarlyStopper(patience=3)
14:    model.train(epochs, optimizer, early_stopper, train_dataset, val_dataset)
15:    best_model_weights  $\leftarrow$  early_stopper
16:    model_initial_weights  $\leftarrow$  best_model_weights
17:  end for
18: end for
```
